TAMIL NADU STATE BOARD - STANDARD 12 CHEMISTRY VOLUME 2

CHAPTER 15: CHEMISTRY IN EVERYDAY LIFE

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 12 English Medium
Subject:	Chemistry Volume 2
Chapter Name:	Chemistry in Everyday Life

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)

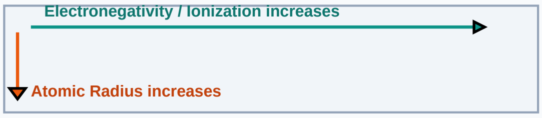
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Antiseptic	Chemical agents applied to living tissues that kill or prevent growth of microorganisms without tissue damage (e.g. Dettol).
Biodegradable Detergent	Surfactants with linear hydrocarbon chains that can be easily decomposed by soil microorganisms.
Food Preservative	Chemical substances added to food to prevent spoilage from microbial growth or oxidation (e.g. sodium benzoate).

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

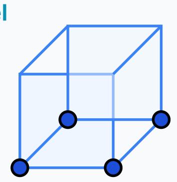
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Simple Cubic Lattice System

Essential Formulas, Laws & Identities:

- Soap Saponification: Glyceryl ester of fatty acid + 3NaOH -> Glycerol + 3Soap
- Aspirin (Acetylsalicylic acid) synthesis: Salicylic acid + Acetic anhydride

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

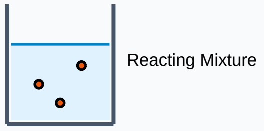
Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Differentiate between antiseptics and disinfectants.

Step-by-step Solution:
Antiseptics are safe for living tissues (e.g., 0.2% phenol, dettol). Disinfectants are toxic for living tissue and applied to non-living floors/drains (e.g., 1% phenol).

Titration Beaker & Solute Precipitation



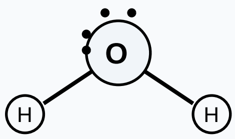
HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Explain the cleansing action of soaps and the formation of micelles in water cleaning.

Analytical Solution Strategy:
Soap molecules have a hydrophilic polar head (-COONa) and a hydrophobic non-polar tail (hydrocarbon).
In water, they organize into spherical micelles: hydrophobic tails cluster inward around oil/dirt, while hydrophilic heads point outward. Rinsing pulls the micelle and trapped dirt away.

Lewis Dot Structure (H_2O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Classify drugs based on therapeutic action with examples.

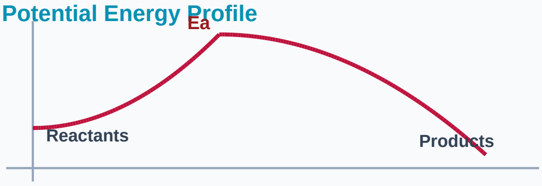
Q2 (2-Mark): What are artificial sweetening agents? Why are they used?

Q3 (2-Mark): Explain saponification of fats.

Q4 (5-Mark): Describe the difference between cationic and anionic detergents.

Q5 (5-Mark): Write a note on thermoplastic and thermosetting polymers.

Chemical Reaction Potential Energy Profile



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Clinical Hygiene

Phenols are diluted to formulate surgical hand-washes to disinfect clinical staff.

Application Case: Food Packaging

Preservatives are added to processed foods to maintain shelf life bounds during shipping.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C_6H_6) Resonance

Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.

✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.

✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3, N) for the final numeric answer.

✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

[ ] Read and memorized core definitions on Page 2.

[ ] Reviewed critical formulas and relations on Page 3.

[ ] Solved worked examples on Page 4.

[ ] Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress. Good luck!

Electrochemical Galvanic Cell System